2. Random Variables (Part II)

Young-geun Kim

Department of Mathematical Sciences

MTH345: Deep Generative Models: A Statistical Perspective

ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Outline

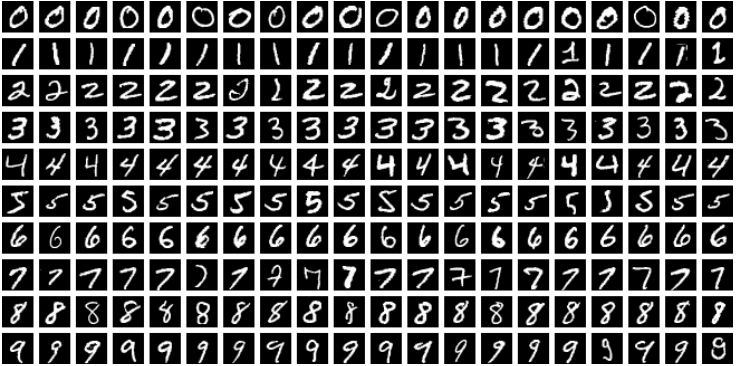
Conditional Distribution

- In probabilistic modeling, we often seek to understand how the distribution of one random variable changes conditionally on another.
- For two continuous random variables X and Y, the conditional probability density function of X given $Y = y$ is defined as:

$$f(x \mid y) = \frac{f(x, y)}{f(y)}$$

  assuming $f(y) > 0$.
- X and Y are independent if and only if $f(x \mid y) = f(x)$ for all x and y.

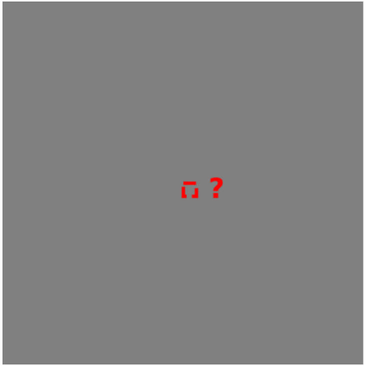
Conditional Distribution



- Consider images from the Modified National Institute of Standards and Technology (MNIST) database (LeCun, 1998).
- Each image consists of $28 \times 28 = 784$ binary pixels, with values in $\{0, 1\}$.

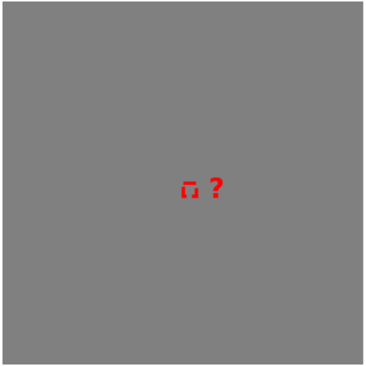
Image source: `https://en.wikipedia.org/wiki/MNIST_database`.

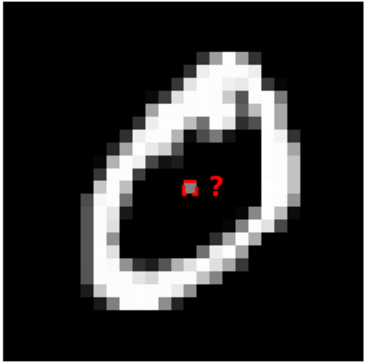
Conditional Distribution



Outcome (x)	0 (black)	1 (white)
$P(X_{14\times14} = x)$	48.08%	51.92%

Conditional Distribution





Outcome (x)	0 (black)	1 (white)
$P(X_{14\times 14} = x \mid \boldsymbol{X}_{-14\times 14})$	48.08% $\to$?	51.92% $\to$?

Bayesian Network

- The *chain rule* can factor any joint PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$ of a random vector $\boldsymbol{X} = (X_1, \ldots, X_d)^T$ into a product of conditional densities.
- For example, when $d = 2$,

$$f_{X_1,X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2|X_1}(x_2 \mid x_1) = f_{X_2}(x_2) f_{X_1|X_2}(x_1 \mid x_2). \tag{1}$$

- As another example, when $d = 3$,

$$\begin{aligned} f_{\boldsymbol{X}}(\boldsymbol{x}) &= f_{X_1}(x_1) f_{X_2|X_1}(x_2 \mid x_1) f_{X_3|X_1,X_2}(x_3 \mid x_1, x_2) \\ &= \cdots \\ &= f_{X_3}(x_3) f_{X_2|X_3}(x_2 \mid x_3) f_{X_1|X_2,X_3}(x_1 \mid x_2, x_3). \end{aligned} \tag{2}$$

- By repeatedly applying the definition of conditional probability:

$$f(x_1, \ldots, x_d) = f(x_1) \prod_{i=2}^{d} f(x_i \mid x_1, \ldots, x_{i-1}).$$

Q: How do we model the dependence within $\boldsymbol{X}$ or the joint PDF $f(x_1, \ldots, x_d)$?

Bayesian Network

1. **Full Model**: Without any domain knowledge, we can model all the conditional distributions,

$$f(x_i \mid x_1, \ldots, x_{i-1}). \tag{3}$$

- It can express any joint distribution, but the model size becomes large.

2. **Degenerate Model**: When $(X_2, \ldots, X_d)$ are known deterministic functions of X_1, modeling $f(X_1)$ alone is sufficient.

- The model size is small and even invariant to the data dimension. However, the model capacity is significantly reduced.

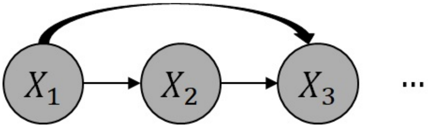
Building appropriate models that reflect domain knowledge is crucial.

Bayesian Network

- *Bayesian networks* are probabilistic graphical models that represent conditional dependencies using directed acyclic graph structures.
- Again, we can express the joint PDF $f(x_1, \ldots, x_d)$ as

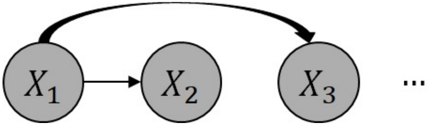
$$f(x_1)f(x_2 \mid x_1) \cdots f(x_d \mid x_1, x_2, \ldots, x_{d-1}). \tag{4}$$

Many dependency structures can be represented by a directed graph with d nodes $X_1, \ldots, X_d$ and up to $d(d-1)/2$ directed edges.

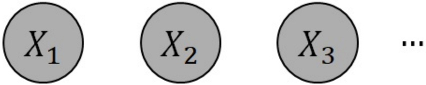


- For example, **1. Full Model** corresponds to a graph utilizing all nodes and all possible directed edges.

Bayesian Network

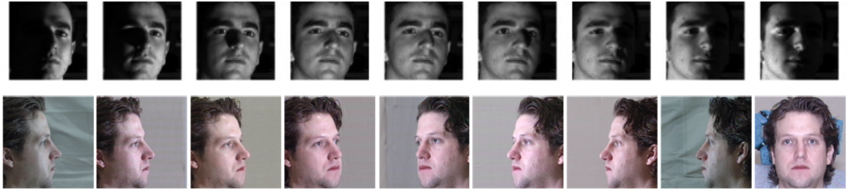


2. **Degenerate Model**: This model is a special case of the above graph where $f(x_1, \ldots, x_d) = f(x_1)f(x_2 \mid x_1) \cdots f(x_d \mid x_1)$.



3. **Fully Independent Model**: This model assumes mutual independence among all variables, such that $f(x_1, \ldots, x_d) = f(x_1) \cdots f(x_d)$.

Bayesian Network: Latent Variable Model

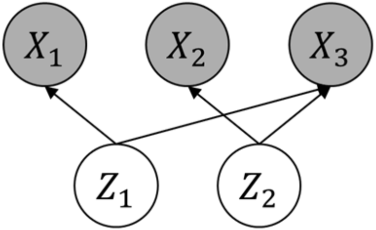


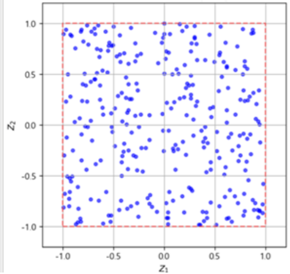
4. **Latent Variable Model**: There are latent factors $\boldsymbol{Z} := (Z_1, \ldots, Z_r)^T$, typically consisting of independent components, that are mixed to generate data, e.g., $\boldsymbol{X} = W\boldsymbol{Z}$ in Independent Component Analysis (Hyvärinen and Oja, 2000).

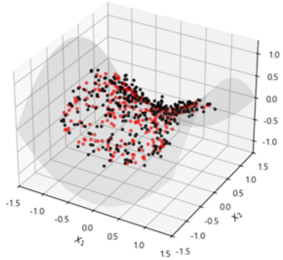
- The age, light source location, and camera angle are examples of latent factors.

The top and bottom images are from the Extended Yale-B (Georghiades et al., 2001) and Multi-pie (Gross et al., 2010) datasets, respectively.

Bayesian Network: Latent Variable Model





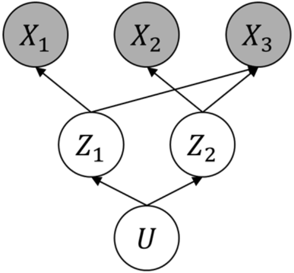


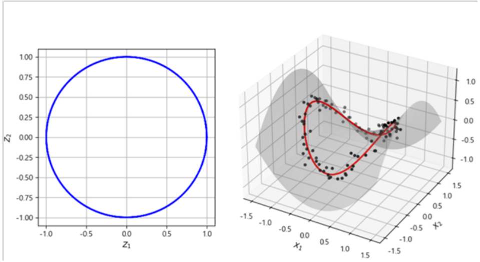
- For example, suppose $\boldsymbol{Z} \sim \mathcal{U}[-1,1] \times \mathcal{U}[-1,1]$ is drawn from a square and $\boldsymbol{X}$ is:

$$X_1 = Z_1 + \epsilon_1, \quad X_2 = Z_2 + \epsilon_2, \quad X_3 = (Z_1^2 - Z_2^2)/2 + \epsilon_3 \tag{5}$$

  where ϵ represents white noise.
- Here, Z_1 and Z_2 are *parent nodes* with no incoming edges, while X_1, X_2, and X_3 are *child nodes* with ones.

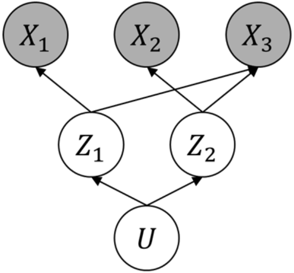
Bayesian Network: Latent Variable Model

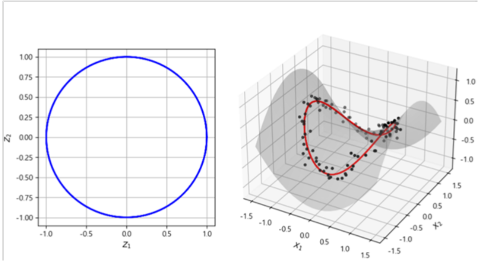




- Now, suppose $\boldsymbol{Z} = (\cos 2\pi U, \sin 2\pi U)^T$ where $U \sim \mathcal{U}[0, 1]$.

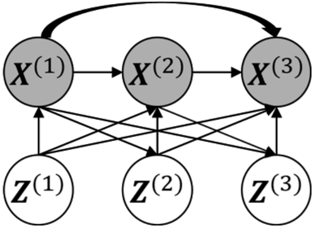
Bayesian Network: Latent Variable Model





- The joint PDF can be expressed as:

Bayesian Network: Latent Variable Model



- The joint PDF can be expressed as:

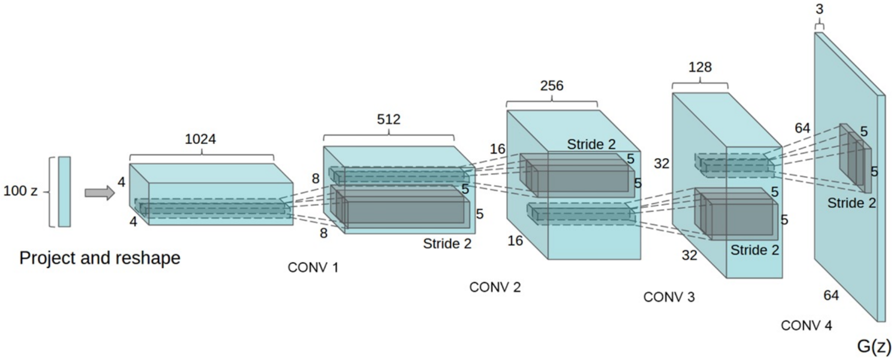
Left images are edited from Szeto et al. (2019).

Bayesian Network: Latent Variable Model

- Joint PDFs from Bayesian networks can be expressed as the product of conditional densities of each node given its parents, denoted as $\mathrm{Pa}(X_i)$:

$$f(x_1, \ldots, x_d) = \prod_{i=1}^{d} f(x_i \mid \mathrm{Pa}(x_i))$$

Bayesian Network: Latent Variable Model



- Deep generative models are predominantly based on latent variable models.
- The conditional distributions $f(\boldsymbol{X}|\boldsymbol{Z})$ can be modeled as parametric distributions, e.g., $N(\boldsymbol{\mu}_{\boldsymbol{X}|\boldsymbol{Z}}(\boldsymbol{Z}), \boldsymbol{\Sigma}_{\boldsymbol{X}|\boldsymbol{Z}}(\boldsymbol{Z}))$.

An example neural network architecture for deep generative model from Radford (2015).

References I

Georghiades, A. S., Belhumeur, P. N., and Kriegman, D. J. (2001). From few to many: Illumination cone models for face recognition under variable lighting and pose. *IEEE transactions on pattern analysis and machine intelligence*, 23(6):643–660.

Gross, R., Matthews, I., Cohn, J., Kanade, T., and Baker, S. (2010). Multi-pie. *Image and vision computing*, 28(5):807–813.

Hyvärinen, A. and Oja, E. (2000). Independent component analysis: algorithms and applications. *Neural networks*, 13(4-5):411–430.

LeCun, Y. (1998). The mnist database of handwritten digits. *http://yann. lecun. com/exdb/mnist/*.

Radford, A. (2015). Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*.

References II

Szeto, R., Sun, X., Lu, K., and Corso, J. J. (2019). A temporally-aware interpolation network for video frame inpainting. *IEEE transactions on pattern analysis and machine intelligence*, 42(5):1053–1068.